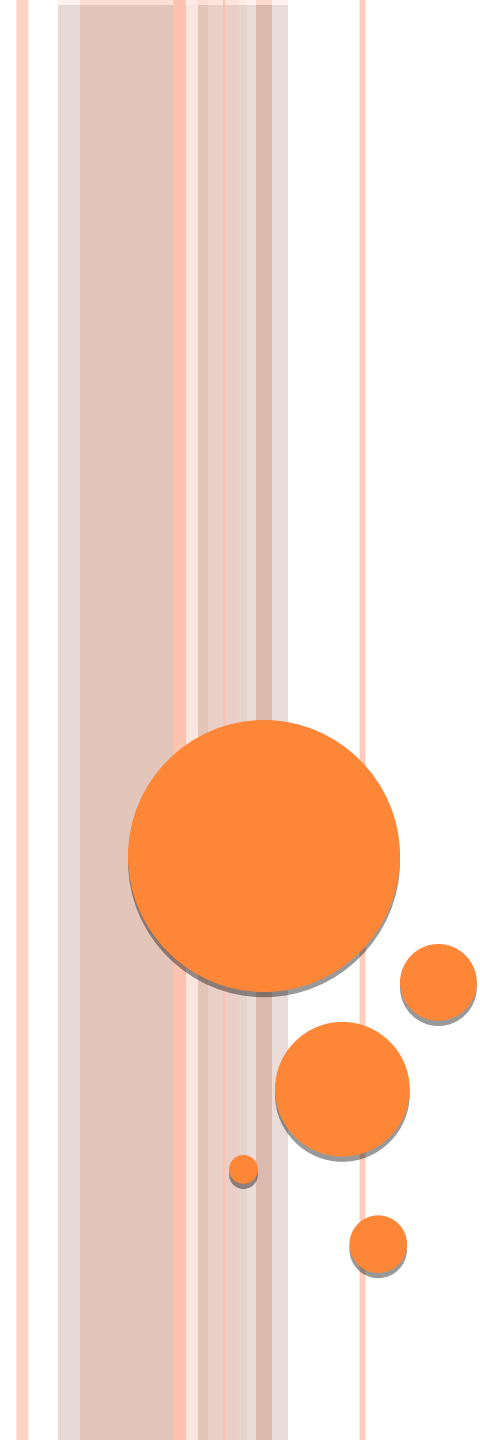




LECTURE 4

STACK

Prepared By:

Afsana Begum
Senior Lecturer,
Software Engineering Department,
Daffodil International University.

CONTENTS

Stack Introduction

Diagram of stacks

Array implementation of stacks

Array Representation of Stacks

Pushing

Popping

Problem and Solution

Quicksort



STACK

A stack is a homogeneous collection of elements in which an element may be inserted or deleted only at one end, called the top of the stack.

Formally this type of stack is called a Last In, First Out (LIFO) stack.

Special terminology is used for two basic operations associated with stacks:

- a) “Push” is the term used to insert an element into a stack.
- b) “Pop” is the term used to delete an element from a stack.



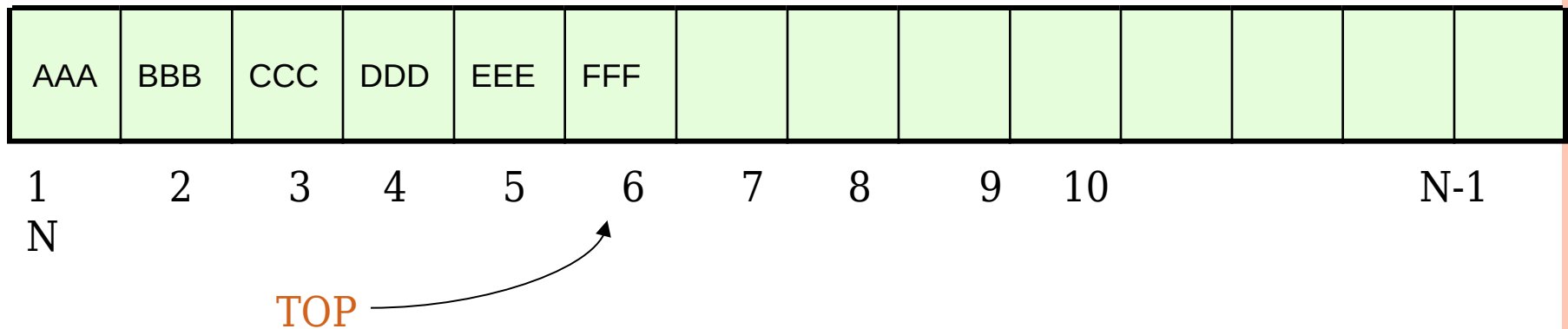
DIAGRAM OF STACKS

Suppose the following 6 elements are pushed, in order, onto an empty stack:

AAA, BBB, CCC, DDD, EEE, FFF

Frequently designate the stack by writing:

STACK : AAA, BBB, CCC, DDD, EEE, FFF



ARRAY IMPLEMENTATION OF STACKS

To implement a stack, items are inserted and removed at the same end (called the **top**)

To use an array to implement a stack, you need both the array itself and an integer

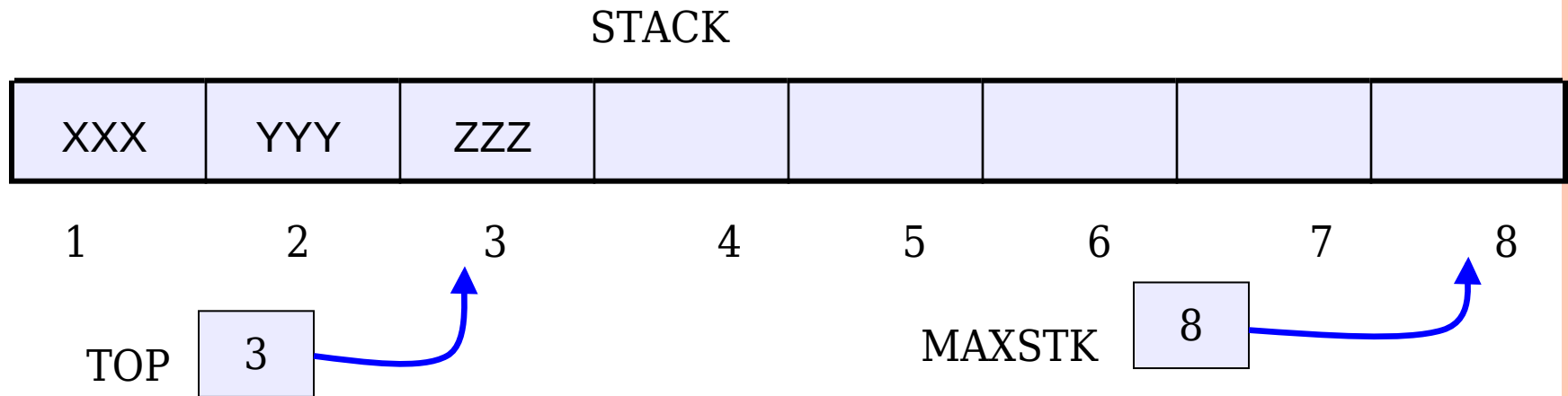
The integer tells you either:

- Which location is currently the top of the stack, or

- How many elements are in the stack



ARRAY REPRESENTATION OF STACKS



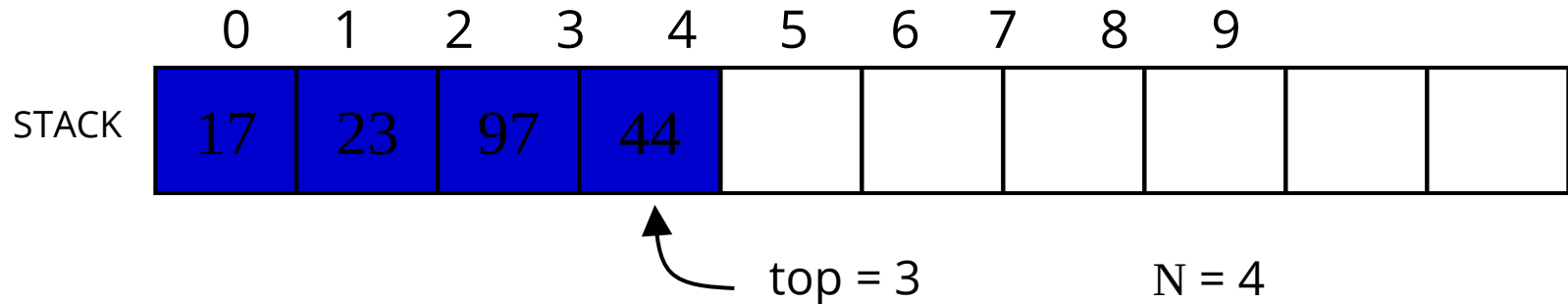
STACK : A linear array

TOP : A pointer variable, Which contains the location of the top element of the stack.

MAXSTK : Gives the maximum number of elements that can be held by the stack.

TOP = 0 or NULL will indicate that the stack is empty

PUSHING

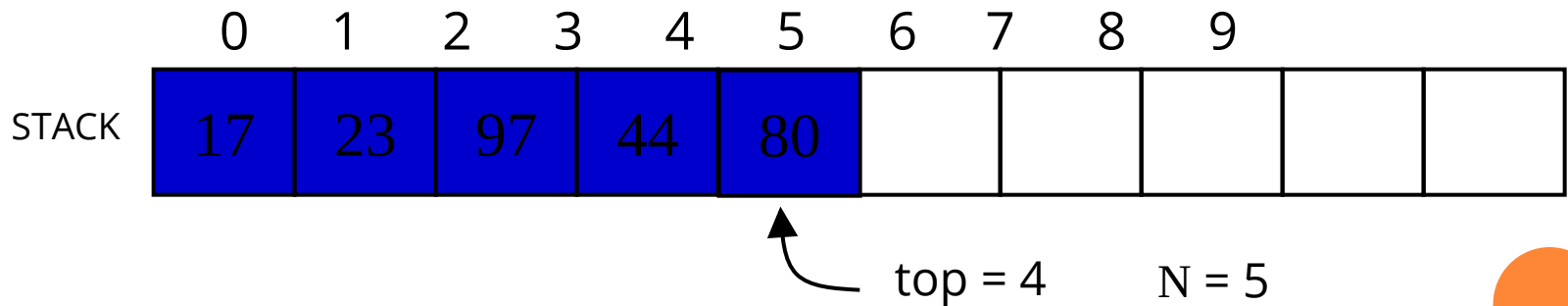


If ITEM=80 is to be inserted, then $TOP = TOP + 1 = 4$

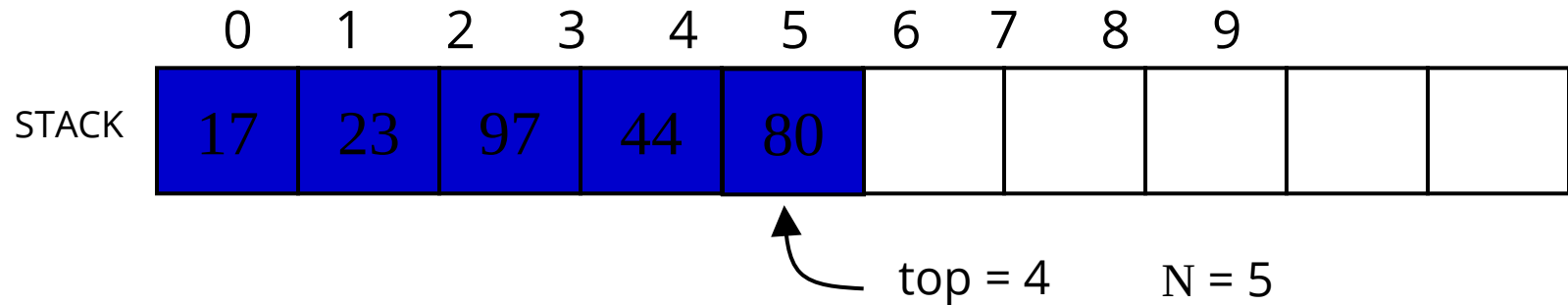
$STACK[TOP] := STACK[4] := ITEM = 80$

N=5

MAXSTACK=10



POPING

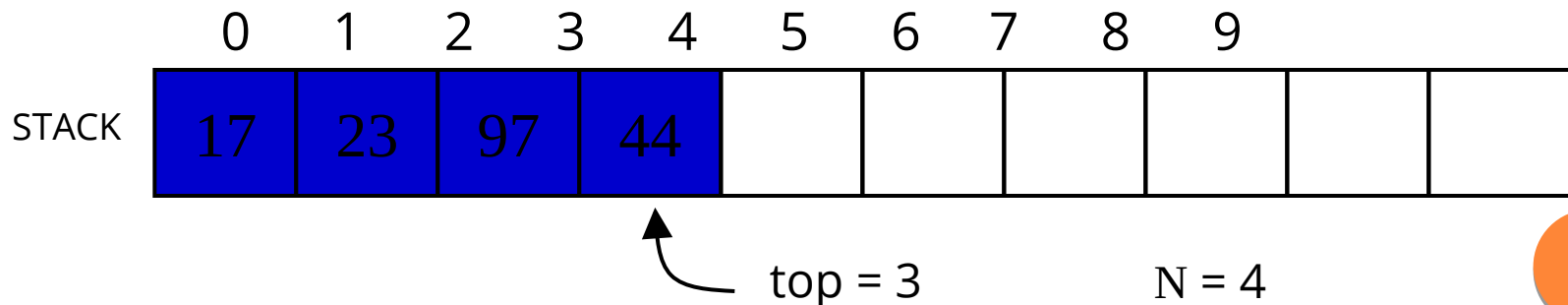


If $\text{ITEM} = 80$ is to be deleted, then $\text{ITEM} := \text{STACK}[\text{TOP}] = \text{STACK}[4] = 80$

$\text{TOP} = \text{TOP} - 1 = 3$

$N = 4$

$\text{MAXSTACK} = 10$



A PROBLEM:

Consider the following stack, where STACK is allocated N=8 memory cells:

STACK: 1,5,3,8,9,0,11,-----

Describe the stack as the following operation take place:

- a) PUSH (STACK , 12)
- b) POP (STACK , ITEM)
- c) PUSH (STACK , 10)
- d) PUSH (STACK , 15)
- e) POP (STACK , ITEM)
- f) PUSH (STACK , 20)
- g) POP (STACK , ITEM)
- h) POP (STACK , ITEM)
- i) POP (STACK , ITEM)
- j) POP (STACK , ITEM)
- K) POP (STACK , ITEM)
- l) POP (STACK , ITEM)
- M) POP (STACK , ITEM)
- N) POP (STACK , ITEM)
- o) POP (STACK , ITEM)



SOLUTION OF THE PROBLEM

Consider the following stack, where STACK is allocated N=8 memory cells:

STACK: 1,5,3,8,9,0,11,-----

Here is the results: (Description of answer is must)

- a) 1,5,3,8,9,0,11,12
- b) 1,5,3,8,9,0,11
- c) 1,5,3,8,9,0,11,10
- d) Overflow , 8 memory cell is used already
- e) 1,5,3,8,9,0,11,
- f) 1,5,3,8,9,0,11,20
- g) 1,5,3,8,9,0,11,
- h) 1,5,3,8,9,0
- i) 1,5,3,8,9
- j) 1,5,3,8
- K) 1,5,3
- l) 1,5
- M) 1
- N) Null
- o) Underflow



STACK SEARCH

Linear Search from top to lower index.

